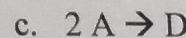
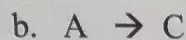


Name: Katrina Mulachy

CHE 321; Spring - 2020; Quiz - 3

100 Points

1. (25 points) Three gas phase reactions have the same reaction rate dependence to the concentration of the reactant.



~~Volume increases, Pressure decreases~~  $P_1 V_1 = P_2 V_2$

Which of these reactions will exhibit the highest conversion in a PFR? Explain.

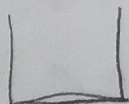
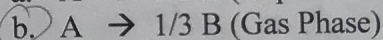
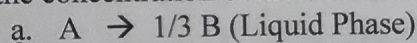
10 ~~b.~~ c. two molecules of A combine into one molecule of D this decreases the pressure which decreases the volume which decreases the concentration of D

2. (25 points) Why does the conversion for a gas-phase reaction in a PBR depend on the pressure?

as pressure drop is significant  
Because the pressure is related to concentration for gases. the A gas adjusts its position

20 to fill the volume of a container, unlike Solids or liquids.  $PV = nRT$ : pressure is related to volume  $\downarrow$   
(incompressible) volume is related to concentration

3. (25 points) Two reactions (one gas phase and one liquid phase) have the same reaction rate dependence to the concentration of the reactant.



Which of these reactions will exhibit the highest conversion in a PFR? Explain.

b. because 3 molecules of A will go to 1 molecule of B. This will cause the concentration of the gas, which is related to the pressure to change. The liquid this does not change.

4. (25 points) A reaction rate equation is given by  $r_A' = k' P_A$ . What are the units of  $k'$ ?

$\frac{\text{mol}}{\text{Pa s kg}}$

$r_A' = \frac{[\text{mol}]}{[\text{kg}][\text{s}]} = k' P_A = k' [\text{atm}]$

$\Rightarrow k' = \frac{[\text{mol}]}{[\text{kg}][\text{s}][\text{atm}]}$